Module 3: ADVANCED MACHINE LEARNING

Course Code: 22DS102006 - Supervised, Unsupervised & Deep Neural Networks

1. Supervised Learning & Kernel Methods

Covers support vector machines (SVM), maximum margin hyperplanes, and the kernel trick (RBF, Polynomial) for non-linear boundaries. Rigorous evaluation using ROC-AUC, Precision-Recall curves, and confusion matrix analytics.

2. Unsupervised Clustering & Density Estimation

Partitioning methods including K-Means, K-Medoids, and silhouette coefficient optimization. Density-based clustering (DBSCAN) for discovering non-spherical clusters and isolating noise/outliers.

3. Deep Neural Architectures

From single-layer perceptrons to deep feed-forward networks (MLP). Activation functions (ReLU, GELU, Softmax), backpropagation calculus via computational graphs, and loss function landscapes (Cross-Entropy, MSE).

4. Convolutional & Sequential Architectures

2D Convolutional neural networks (CNN) with pooling layers for spatial computer vision. Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) cells for temporal sequential modeling.